

Second order homogeneous ODEs

Basic skills

Q1. a) $x^2 - 5x + 6 = 0$

$(x-3)(x-2) = 0$ so $x = 2$ or 3
 $\uparrow$
 2 real

b) $x^2 - 8x + 16 = 0$

$(x-4)(x-4) = 0$ $x = 4$ (or 4)
 $\uparrow$
 1 real

c) $x^2 - 8x + 20 = 0$

$x = \frac{8 \pm \sqrt{64 - 80}}{2} = 4 \pm 2i$ ← 2 complex

Q2. a) $\frac{d^2y}{dx^2} + y = 0$

$\uparrow$ constants 1, 1 $\uparrow$ homo
 $\downarrow$ $\downarrow$ In homo

- homogeneous as no $f(x)$ alone
- Constant coefficients

b) $\frac{d^2y}{dx^2} + y = -x$

constants $\uparrow$ In homo

- NOT homogeneous as $f(x)$ alone
- Constant coefficients
- NOT homogeneous as $f(x)$ alone
- Constant coefficients

c) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \cos x$

$$d) \quad x \frac{d^2 y}{dx^2} + 3x^2 \frac{dy}{dx} + 2x^3 y = 0$$

- Homogeneous as no $f(x)$ alone
- Not constant coefficients as
 $x, 3x^2, 2x^3$

$$Q3. \quad 2 \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = 0$$

y'' is m^2



y' is m



y is m^0

$$2m^2 - 4m + 1 = 0$$

$$Q4. \quad y = 3e^{2x} \Rightarrow y' = 6e^{2x} \Rightarrow y'' = 12e^{2x}$$

$$\text{so in } \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y$$

$$\Rightarrow 12e^{2x} - 24e^{2x} + 12e^{2x} = 0e^{2x} = 0$$

so yes it does equal zero and

hence is a solution

Competency skills

Q5. $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

a) $m^2 - 5m + 6 = 0 \Rightarrow (m-3)(m-2) = 0$

$\Rightarrow m = 2 \text{ or } 3$

b) As two real have

$y = Ae^{2x} + Be^{3x}$

(2 real
 $y = Ae^{m_1x} + Be^{m_2x}$)

c) Apply $y(0) = 2 \Rightarrow Ae^0 + Be^0 = 2$
 $\Rightarrow A + B = 2$

For $\frac{dy}{dx}(0) = 5$ need $\frac{dy}{dx} = 2Ae^{2x} + 3Be^{3x}$

then $2Ae^0 + 3Be^0 = 5 \Rightarrow 2A + 3B = 5$

Solving $A + B = 2, 2A + 3B = 5 \Rightarrow B = 1$
 $\Rightarrow A = 1$

So Particular solution

$y = e^{2x} + e^{3x}$

Q6. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

so auxiliary: $m^2 + 4m + 4 = 0$

$\Rightarrow (m+2)^2 = 0 \Rightarrow m = -2$

as repeated root

$y = Ae^{-2x} + Bxe^{-2x}$

(1 root)
 $y = Ae^{mx} + Bxe^{mx}$

Q7. $9\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0$ $y(0) = 0$
 $y(1) = 2$

auxiliary: $9m^2 - 12m + 4 = 0$

$(3m-2)(3m-2) = 0$ $m = \frac{2}{3}$

so $y = Ae^{\frac{2}{3}x} + Bxe^{\frac{2}{3}x}$

Then $Ae^0 + B \times 0 \times e^0 = 0 \Rightarrow A = 0$

so $y = Bxe^{\frac{2}{3}x}$ so $B \times 1 \times e^{\frac{2}{3}} = 2$

$\Rightarrow B = 2e^{-\frac{2}{3}}$ so $y = 2e^{-\frac{2}{3}}xe^{\frac{2}{3}x}$
 $= 2xe^{\frac{2}{3}(x-1)}$

Q8. $\frac{2}{x} \frac{d^2y}{dx^2} + \frac{4}{x} \frac{dy}{dx} + \frac{3y}{x} = 0$ with $y(0)=0$,
 $\Rightarrow 2 \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 3y = 0$ $\frac{dy}{dx}(0)=1$

auxiliary $2m^2 + 4m + 3 = 0$

so $m = \frac{-4 \pm \sqrt{16 - 24}}{4} = \frac{-4 \pm i\sqrt{8}}{4}$
 $= -1 \pm \frac{1}{2}\sqrt{2}$

so $y = e^{-1x} \left(A \cos\left(\frac{1}{2}\sqrt{2}x\right) + B \sin\left(\frac{1}{2}\sqrt{2}x\right) \right)$

Then $y(0)=0 \Rightarrow 1 \times (A \times 1 + B \times 0) = 0$
 $\Rightarrow A = 0$

so $y = e^{-x} B \sin\left(\frac{1}{2}\sqrt{2}x\right)$

need $\frac{dy}{dx} = e^{-x} \frac{B\sqrt{2}}{2} \cos\left(\frac{1}{2}\sqrt{2}x\right) - e^{-x} B \sin\left(\frac{1}{2}\sqrt{2}x\right)$
↓ product rule

so $\frac{dy}{dx}(0)=1 \Rightarrow \frac{B\sqrt{2}}{2} = 1 \Rightarrow B = \frac{2}{\sqrt{2}} = \sqrt{2}$

so $y = \sqrt{2} e^{-x} \sin\left(\frac{1}{2}\sqrt{2}x\right)$

Advanced skills

decreasing of rate

Q9.

a) By statement $\frac{dT}{dt} = -3 \frac{d^2T}{dt^2}$

$$\Rightarrow 3 \frac{d^2T}{dt^2} + \frac{dT}{dt} = 0$$

b) $T(0) = 30$ $T(2) = 20$

auxiliary $3m^2 + m = 0 \Rightarrow m(3m+1) = 0$

$$\Rightarrow m = 0 \text{ or } m = -\frac{1}{3} \text{ so}$$

$$T = Ae^{0t} + Be^{-\frac{1}{3}t} = A + Be^{-\frac{1}{3}t}$$

Applying conditions $A + B = 30$

$$A + Be^{-\frac{2}{3}} = 20$$

$$\text{so } B(1 - e^{-\frac{2}{3}}) = 10 \quad B = \frac{10}{1 - e^{-\frac{2}{3}}}$$

$$\begin{aligned} \text{so } A &= 30 - \frac{10}{1 - e^{-\frac{2}{3}}} \\ &= \frac{20 - 30e^{-\frac{2}{3}}}{1 - e^{-\frac{2}{3}}} \end{aligned} \Rightarrow T = \frac{20 - 30e^{-\frac{2}{3}} + 10e^{-\frac{2}{3}t}}{1 - e^{-\frac{2}{3}}}$$

$$\text{so } T(s) = \frac{20 - 30e^{-\frac{2}{3}} + 10e^{-\frac{10}{3}}}{1 - e^{-\frac{2}{3}}}$$

$$\approx 10.18$$

$$\text{c) As } t \rightarrow \infty \quad e^{-\frac{2}{3}t} \rightarrow 0$$

$$\text{so } T \rightarrow \frac{20 - 30e^{-\frac{2}{3}}}{1 - e^{-\frac{2}{3}}} = 9.449$$

$$\text{Q10. } 5 \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 5y = 0$$

$$5m^2 - 2m + 5 = 0$$

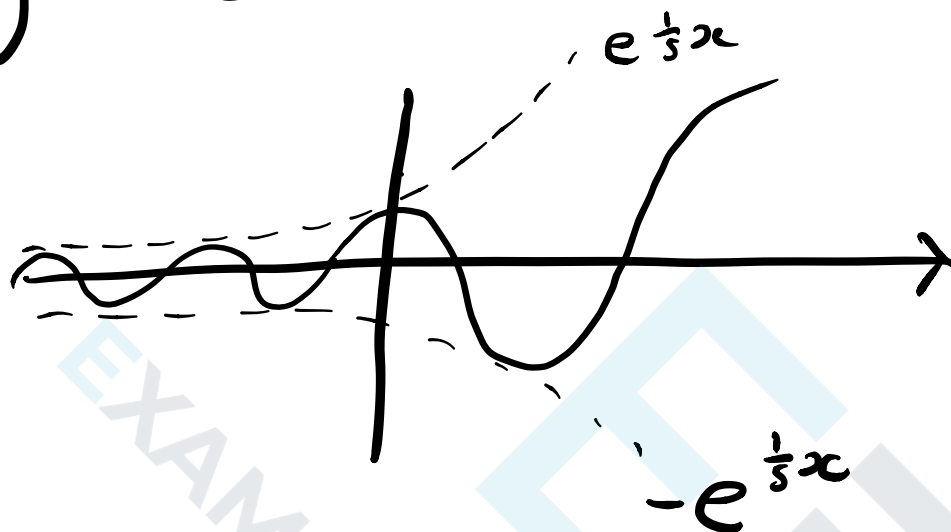
$$m = \frac{2 \pm \sqrt{4 - 100}}{10} = \frac{2 \pm i\sqrt{96}}{10}$$

$$= \frac{1}{5} \pm \frac{2}{5} \sqrt{6} i$$

$$y = e^{\frac{1}{5}x} \left(A \cos\left(\frac{2}{5}\sqrt{6}x\right) + B \sin\left(\frac{2}{5}\sqrt{6}x\right) \right)$$

gle

$$y = C e^{\frac{1}{5}x} \cos\left(\frac{2}{5}\sqrt{x} + D\right)$$



For $C, D \neq 0$